Project: Worked examples
Title: Member checks (EN 1993-1-1)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the resistance of a class 1, 2 or 3 steel member to tension, compression, buckling, bending, shear and their interaction, to EN 1993-1-1. Tension is positive; class 4 sections and lateral-torsional buckling are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\ f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\ f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\ f_y/(\sqrt{3}\ \gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\ V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$. Class 1–2 I-section with h, t_w, t_f entered: $M_{y,V,Rd} = (W_{pl,y} - \rho\ A_w^2/(4\ t_w))\ f_y/\gamma_{M0}$, $A_w = (h - 2t_f)\ t_w$, and $M_{z,c,Rd}$ is not reduced; otherwise the whole resistance × $(1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\ f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\ A\ f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling (6.61, 6.62)	$N_{Ed}/N_{b,Rd,min} + k_{yy}\ M_{y,Ed}/(W_y\ f_y/\gamma_{M1}) + k_{zz}$ $M_{z,Ed}/(W_z\ f_y/\gamma_{M1}) \le 1$, the characteristic bending resistance M_{Rk} over γ_{M1} without the shear reduction

Basis: Tension[1](cl. 6.2.3); Compression[1](cl. 6.2.4); Bending moment[1](cl. 6.2.5); Shear[1](cl. 6.2.6); Bending and shear[1](cl. 6.2.8); Linear summation of utilisations[1](cl. 6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](cl. 6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](cl. 6.3.3); Imperfection factors for the buckling curves[2](§6.3.1, 1st ed.).

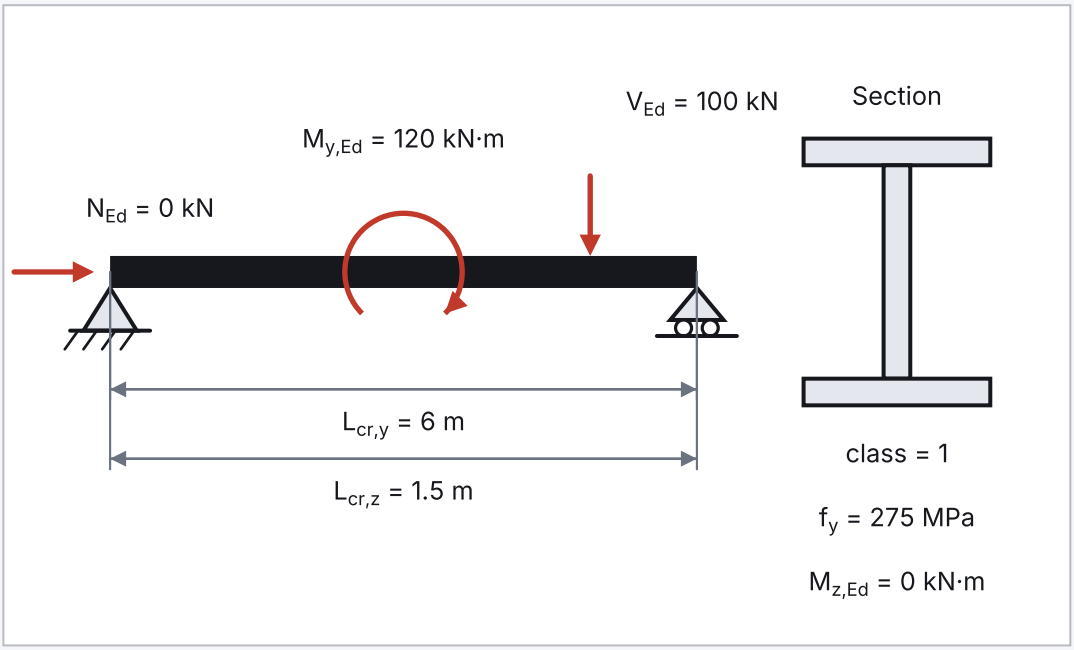


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$section_{class}$	Cross-section class	1		assumption: UB 305×165×40 S275 is class 1 in bending (Blue Book).	any
$curve_{y}$	Buckling curve about y-y	a		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, y-y	any
$curve_{z}$	Buckling curve about z-z	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, z-z	any
ltb	Compression flange against lateral-torsional buckling (optional; blank = blocks when $M_{y,Ed}$ is above 0)	Restrained		assumption: The deck restrains the compression flange (at 1.5 m, as L_cr_z), so lateral-torsional buckling does not govern; 6.3.2 is outside this sheet.	any
A	Area	51.3	cm^2	reference: S1 section sheet / Blue Book UB 305×165×40	1 … 100000 cm^2
$W_{pl,y}$	Plastic section modulus about y-y	623	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,y}$	Elastic section modulus about y-y	560	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$W_{pl,z}$	Plastic section modulus about z-z	142	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,z}$	Elastic section modulus about z-z	92.6	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	Second moment of area about y-y	8503	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	Second moment of area about z-z	764	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth for the shear area (optional; blank = use A_v)	303.4	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
b	Width for the shear area (optional; blank = use A_v)	165	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
t_w	Web thickness for the shear area (optional; blank = use A_v)	6	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
t_f	Flange thickness for the shear area (optional; blank = use A_v)	10.2	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
r	Root radius for the shear area (optional; blank = 0)	8.9	mm	spec: BS EN 10365 UB 305×165×40	0 … 500 mm
f_y	Yield strength for the thickness	275	MPa	spec: BS EN 10025-2 S275, t ≤ 16 mm	200 … 700 MPa
E	Young's modulus	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
N_{Ed}	Design axial force, tension positive and compression negative, opposite in sign to the base plate sheet's axial force	0	kN	reference: S2 frame sheet, two-span beam, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major axis)	120	kN·m	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor axis)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	100	kN	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN
$L_{cr,y}$	Buckling length about y-y	6	m	assumption: Span 6 m.	0.1 … 100 m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$L_{cr,z}$	Buckling length about z-z	1.5	m	assumption: Compression flange restrained at 1.5 m by the deck.	0.1 … 100 m
γ_{M0}	Partial factor for cross-section resistance	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor for member buckling	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M2}	Partial factor for net-section fracture	1.1		default	1 … 1.35
η	Shear-area factor	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
k_{yy}	Interaction factor for bending about y-y in the member check	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2
k_{zz}	Interaction factor for bending about z-z in the member check	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2

3. Assumptions

A1. The cross-section class is the class the user states, 1, 2 or 3. (Class 4 effective properties are not derived here.)
A2. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Lateral-torsional buckling of a member that is not restrained is checked on the LTB interaction sheet.)
A3. A k left at 1.0 is replaced by the Annex A or B upper bound $C_m(1 + 0.8n)$. (Applies under compression with a moment; the smaller flexural-buckling reduction is used on the axial term and the factors are not derived.)
A4. Imperfection factors are the textbook values for the buckling curve unless the user overrides them. (Gardner and Nethercot, Designers' Guide to EN 1993-1-1.)
A5. Partial factors are 1.0 as in the UK National Annex unless the user overrides them. (UK National Annex to BS EN 1993-1-1.)
A6. $\text{section}_{\text{class}}$ (Cross-section class) is taken as 1. (UB 305×165×40 S275 is class 1 in bending (Blue Book).)
A7. ltb (Compression flange against lateral-torsional buckling (optional; blank = blocks when $M_{y,Ed}$ is above 0)) is taken as Restrained. (The deck restrains the compression flange (at 1.5 m, as $L_{cr,z}$), so lateral-torsional buckling does not govern; 6.3.2 is outside this sheet.)
A8. $L_{cr,y}$ (Buckling length about y-y) is taken as 6 m. (Span 6 m.)
A9. $L_{cr,z}$ (Buckling length about z-z) is taken as 1.5 m. (Compression flange restrained at 1.5 m by the deck.)
A10. k_{yy} (Interaction factor for bending about y-y in the member check) is taken as 1. (No axial force; interaction factors irrelevant.)
A11. k_{zz} (Interaction factor for bending about z-z in the member check) is taken as 1. (No axial force; interaction factors irrelevant.)
A12. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Input ltb.)

4. Working

[28] $r_h = r$
$= 8.9\ \text{mm}$
$= 8.9\ \text{mm}$
Root radius used by the helper

[29] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [3]
$= 51.3\ \text{cm}^2 - 2 \times 165\ \text{mm} \times 10.2\ \text{mm} + (6\ \text{mm} + 2 \times 8.9\ \text{mm}) \times 10.2\ \text{mm}$
$= 20.068\ \text{cm}^2$
Rolled I shear area, load parallel to the web

[30] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [3]
$= 1 \times (303.4\ \text{mm} - 2 \times 10.2\ \text{mm}) \times 6\ \text{mm}$
$= 16.98\ \text{cm}^2$
Lower bound $\eta\ h_w\ t_w$

[31] $A_v = \max(A_{v,rolled}, A_{v,min})$
$= \max(20.068\ \text{cm}^2, 16.98\ \text{cm}^2)$
$= 20.068\ \text{cm}^2$
Shear area

[32] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [4](6.2.4(2))
$= 51.3\ \text{cm}^2 \times 275\ \text{MPa} / 1$
$= 1410.8\ \text{kN}$
Plastic axial resistance

[33] $W_y = W_{pl,y}$ [5]
$= 623\ \text{cm}^3$
$= 623\ \text{cm}^3$
Section modulus about y (plastic, class 1–2)

[34] $W_z = W_{pl,z}$ [5]
$= 142\ \text{cm}^3$
$= 142\ \text{cm}^3$
Section modulus about z (plastic)

[35] $V_{pl,Rd} = A_v * f_y / (\text{sqrt}(3) * \gamma_{M0})$ [6]
$= 20.068\ \text{cm}^2 \times 275\ \text{MPa} / (\text{sqrt}(3) \times 1)$
$= 318.62\ \text{kN}$
Plastic shear resistance

[36] $u_V = V_{Ed} / V_{pl,Rd}$
$= 100\ \text{kN} / 318.62\ \text{kN}$
$= 0.31386$
Shear utilisation

[37] $\rho_{shear} = 0$ [7]
$= 0$
Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)

[38] $M_{y,c,Rd} = (1 - \rho_{shear}) * W_y * f_y / \gamma_{M0}$ [8](6.2.8)
$= (1 - 0) \times 623\ \text{cm}^3 \times 275\ \text{MPa} / 1$
$= 171.32\ \text{kN·m}$
Bending resistance about y

[39] $M_{z,c,Rd} = (1 - \rho_{shear}) * W_z * f_y / \gamma_{M0}$ [5]
$= (1 - 0) \times 142\ \text{cm}^3 \times 275\ \text{MPa} / 1$
$= 39.05\ \text{kN·m}$
Bending resistance about z

[40] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$
$= 120\ \text{kN·m} / 171.32\ \text{kN·m}$
$= 0.70042$
Bending utilisation, y

[41] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$
$= 0\ kN·m / 39.05\ kN·m$
$= 0$
Bending utilisation, z
[42] $N_{abs} = abs(N_{Ed})$
$= abs(0\ kN)$
$= 0\ kN$
Axial force magnitude
[43] $u_N = N_{abs} / N_{pl,Rd}$
$= 0\ kN / 1410.8\ kN$
$= 0$
Tension utilisation
[44] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [9]
$= 0 + 0.70042 + 0$
$= 0.70042$
Linear cross-section interaction
[47] α_y = imperfection factor, curve a [10]
= Designers' Guide §6.3.1
= 0.21
Imperfection factor about y
[48] α_z = imperfection factor, curve b [10]
= Designers' Guide §6.3.1
= 0.34
Imperfection factor about z
[49] $N_{cr,y} = pi^2 * E * I_y / L_{cr,y}^2$ [11]
$= π^2 × 210\ GPa × 8503\ cm^4 / (6\ m)^2$
$= 4895.4\ kN$
Elastic critical force about y
[50] $\lambda_y = sqrt(A * f_y / N_{cr,y})$ [11]
$= sqrt(51.3\ cm^2 × 275\ MPa / 4895.4\ kN)$
$= 0.53682$
Non-dimensional slenderness about y
[51] $\Phi_y = 0.5 * (1 + \alpha_y * (\lambda_y - 0.2) + \lambda_y^2)$ [11]
$= 0.5 × (1 + 0.21 × (0.53682 - 0.2) + 0.53682^2)$
$= 0.67946$
Φ about y
[52] $\chi_y = min(1, 1 / (\Phi_y + sqrt(\Phi_y^2 - \lambda_y^2)))$ [11]
$= min(1, 1 / (0.67946 + sqrt(0.67946^2 - 0.53682^2)))$
$= 0.91244$
Reduction factor χ about y
[53] $N_{b,y,Rd} = \chi_y * A * f_y / \gamma_{M1}$ [12]
$= 0.91244 × 51.3\ cm^2 × 275\ MPa / 1$
$= 1287.2\ kN$
Buckling resistance about y
[54] $N_{cr,z} = pi^2 * E * I_z / L_{cr,z}^2$ [11]
$= π^2 × 210\ GPa × 764\ cm^4 / (1.5\ m)^2$
$= 7037.7\ kN$
Elastic critical force about z
[55] $\lambda_z = sqrt(A * f_y / N_{cr,z})$ [11]
$= sqrt(51.3\ cm^2 × 275\ MPa / 7037.7\ kN)$
$= 0.44772$
Non-dimensional slenderness about z

[56] $\Phi_z = 0.5 * (1 + \alpha_z * (\lambda_z - 0.2) + \lambda_z^2)$ [11]
$= 0.5 \times (1 + 0.34 \times (0.44772 - 0.2) + 0.44772^2)$
$= 0.64234$
Φ about z
[57] $\chi_z = \min(1, 1 / (\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}))$ [11]
$= \min(1, 1 / (0.64234 + \sqrt{0.64234^2 - 0.44772^2}))$
$= 0.90667$
Reduction factor χ about z
[58] $N_{b,z,Rd} = \chi_z * A * f_y / \gamma_{M1}$ [12]
$= 0.90667 \times 51.3\ cm^2 \times 275\ MPa / 1$
$= 1279.1\ kN$
Buckling resistance about z
[59] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$
$= \min(1287.2\ kN, 1279.1\ kN)$
$= 1279.1\ kN$
Governing buckling resistance
[60] $u_{buckling} = 0$ [13]
$= 0$
Flexural buckling utilisation (tension: not applicable)
[61] $M_{y,b,Rd} = W_y * f_y / \gamma_{M1}$ [14](6.61)
$= 623\ cm^3 \times 275\ MPa / 1$
$= 171.32\ kN \cdot m$
Characteristic bending resistance about y over γ_{M1} (member check, no shear reduction)
[62] $M_{z,b,Rd} = W_z * f_y / \gamma_{M1}$ [14](6.62)
$= 142\ cm^3 \times 275\ MPa / 1$
$= 39.05\ kN \cdot m$
Characteristic bending resistance about z over γ_{M1} (member check, no shear reduction)
[63] $u_{member} = u_{M,y} + u_{M,z}$ [15]
$= 0.70042 + 0$
$= 0.70042$
Simplified member interaction
[64] $u_{limit} = limit$
$= the\ utilisation\ limit$
$= 1$
Utilisation limit

5. Results

SYMBOL	VALUE	DESCRIPTION
$N_{pl,Rd}$	1410.8 kN	Plastic axial resistance
$N_{b,Rd}$	1279.1 kN	Governing buckling resistance
$M_{y,c,Rd}$	171.32 kN·m	Bending resistance about y-y
$M_{z,c,Rd}$	39.05 kN·m	Bending resistance about z-z
$V_{pl,Rd}$	318.62 kN	Plastic shear resistance
ρ_{shear}	0	Shear reduction factor ρ
$u_{section}$	0.70	Cross-section interaction
$u_{buckling}$	0.00	Buckling utilisation
u_{member}	0.70	Member interaction
A_v	20.07 cm²	Shear area

SYMBOL	VALUE	DESCRIPTION
$M_{y,b,Rd}$	171.32 kN·m	Characteristic bending resistance about y-y over the partial factor
$M_{z,b,Rd}$	39.05 kN·m	Characteristic bending resistance about z-z over the partial factor
$N_{cr,y}$	4895.4 kN	Elastic critical force about y-y
$N_{cr,z}$	7037.7 kN	Elastic critical force about z-z
λ_y	0.53682	Non-dimensional slenderness about y-y
λ_z	0.44772	Non-dimensional slenderness about z-z
χ_y	0.91244	Buckling reduction factor about y-y
χ_z	0.90667	Buckling reduction factor about z-z
$N_{b,y,Rd}$	1287.2 kN	Buckling resistance about y-y
$N_{b,z,Rd}$	1279.1 kN	Buckling resistance about z-z
u_N	0.00	Axial utilisation
$u_{M,y}$	0.70	Bending y-y utilisation (governing)
$u_{M,z}$	0.00	Bending z-z utilisation
u_V	0.31	Shear utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bending moment y-y against bending resistance	120 kN·m	171.32 kN·m	0.70	[16](6.2.5)
PASS	Cross-section interaction against its limit	0.70042	1	0.70	[16](6.2.1(7))
PASS	Shear force against plastic shear resistance	100 kN	318.62 kN	0.31	[16](6.2.6)
PASS	Tension against plastic resistance	0 kN	1410.8 kN	0.00	[17](6.2.4)
PASS	Bending moment z-z against bending resistance	0 kN·m	39.05 kN·m	0.00	[16](6.2.5)

7. Applicability

None.

Defaults used (inputs left empty):
γ_{M2}: not entered, so the default of 1.1 is used.

8. Verification

Template struct.en1993-member@1.6.0 Content hash 09dec90614ac2f64 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.
3. 6.2.6(3).
4. 6.2.3(2),.
5. 6.2.5(2).
6. 6.2.6(2).
7. 6.2.8(2).
8. 6.2.5(2),.
9. 6.2.1(7).
10. 6.3.1.2.
11. 6.3.1.2(1).
12. 6.3.1.1(3).
13. 6.3.1.1(1).
14. 6.3.3, Eq.
15. 6.3.3(4).
16. cl.
17. cl. 6.2.3,.